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RESEARCH ARTICLE

Lightweight End-to-End Patch-Based Self-Attention Network for Robust Image Forgery Detection

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ABSTRACT Image tampering detection and localization have become critical tasks in digital forensics due to the widespread accessibility of powerful image editing tools. We came up with a new Vision Transformer (ViT)-based design that uses patch embedding and multi-head self-attention mechanisms to efficiently find fake documents. This is because we need lightweight but highly accurate detection systems that can model all of a document's features. Our model uses a modular pipeline made up of a patch embedding layer, four transformer encoder blocks, and a streamlined classification head. The slimmed-down classification head consisted of only 69,793 trainable parameters to keep the computation costs as minimal as possible. Experimental evaluation conducted on four benchmark datasets—CASIA v1, DEFACTO (Splicing), MICC-F2000, and Columbia—demonstrates the superior performance of the proposed model. The model gets an AUC of 0.9712 and an F1 score of 0.9587 on CASIA v1, an almost perfect AUC of 0.9993 and an F1 score of 0.9914 on DEFACTO, a good AUC of 0.8491 and an F1 score of 0.8319 on Columbia, and an exeptional AUC of 0.9804 and an F1 score of 0.9097 on MICC-F2000. The proposed architecture was evaluated against 59 different models and demonstrated its effectiveness, showing its strong potential for practical applications such as detecting and localizing document forgeries.

INDEX TERMS Image tampering detection, lightweight deep learning model, vision transformer, self-attention mechanism, patch embedding.

I. INTRODUCTION

Image tampering detection and localization has become an essential field in digital forensics, focused on finding and recognizing tampered areas inside images [1]. The quick development of image editing tools and AI-generated forgeries have made proof of content authenticity progressively harder. Original methods depended on handcrafted features and statistical analysis, but modern techniques include deep learning and transformer-based architecture [2]. These models try to not only recognize forgeries but also exactly map the manipulated areas at the pixel level. This literature study identifies and examines major developments from standard to modern designs. In contrast to existing ViT-based models that often involve complex architectures or high parameter counts, our approach introduces a lightweight, end-to-end design with only 69,793 trainable parameters, enabling efficient and scalable image forgery detection.

Initial approaches to image forgery detection primarily relied on handcrafted features and statistical analysis. One method [3] applied statistical properties obtained from block-based DCT coefficients for recognizing splicing and

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copy-move forgeries, proving precise on JPEG images and efficient, though weak in localization and generalizability to GAN-generated images. Another approach [4] used Discrete Wavelet Transform (DWT) and Dominant Rotated Local Binary Pattern (DRLBP) to identify edge-based textural differences, achieving enhanced accuracy but limited by a fixed descriptor pipeline and lack of localization. A further study [5] improved texture-based detection with OELTP descriptors and energy-driven dimensionality reduction, yet faced difficulties due to grayscale restrictions and lacked advanced deep learning abilities. Additionally, a model [6] mixed Markov features with LBP and DWT for greater performance, but applied static grayscale thresholds and demonstrated poor localization.

These models progressed beyond handcrafted features and introduced early deep learning architectures. One approach [7] proposed a modular design combining CLAHE, a deep CNN, and SVM, displaying great accuracy but struggling with splicing localization and requiring a fixed input size. Another study [8] upgraded AlexNet with Maxout and Batch Normalization for forgery detection and localization, achieving greater cross-dataset accuracy, though it used obsolete architectures and lacked benchmarking with current models. Additionally, a method [9] applied Combined Vision Transformers (ViT) for classification when paired with SAM for localization, offering stable performance but being highly computational and dependent on SAM fine-tuning.

These models focused on pixel-wise tampering localization with encoder-decoder structures. One design [10] introduced GSCNet with Gram blocks and ConvLSTMs to ensure semantic consistency, achieving accuracy but demanding significant memory. Another model [11] implemented multi-scale focused learning and CBAM attention modules for boundary recognition, though performance decreased with JPEG compression. A dual-encoder D-Net [12] excelled in multi-scale localization for fingerprint extraction but was limited to copy-move forgeries. Another approach [11] applied ConvNeXt with multi-scale integration, obtaining superior results but facing issues with tampering regions and requiring significant computation. A lightweight Form model [13] utilized Vision Mamba for global modeling, reducing FLOPs but performing poorly in semantic segmentation and deepfake evaluation.

These methods integrated attention mechanisms and frequency-domain insights. The CANF model [14] combined five noise properties with channel and frequency attention, offering strong and modular performance but being resource-intensive. A self-adversarial training pipeline [15] included a forgery attention module, improving durability and limited detection, though it relied on stationary, handcrafted filters. TANet [15] merged CNNs and Transformers with edge-refinement blocks (BAB, RAMB), achieving better boundary precision at the cost of complexity. ReLoc [16] merged restoration and localization modules for enhanced post-processing performance, though it suffered from a high inference period.

State-of-the-art models leverage transformers and prompt tuning. A PVT-based model [17] included boundary awareness and multi-scale enhancements, delivering excellent performance but struggling with copy-move forgery. SSR-IFL [18] targeted stereo SR forgeries with MDSA and BPFF modules, but stereo data limited real-world generalization. A prompt-tuning method [19] using a frozen Swin-Transformer achieved high generalization across datasets with low training cost, though it performed poorly on copy-move datasets and required high-resolution images. UGEE-Net [20] integrated Bayesian uncertainty with frequency-domain edge enhancement, proving highly efficient for minor manipulations, despite being complicated and unsuitable for real-time applications.

These models explore alternative approaches like disentanglement and perceptual hashing. DiRLoc [21] used a two-stage disentanglement method for effective JPEG compression control, though it required resource-intensive training and mobile optimization. SmartHash [22] introduced a simple perceptual hashing architecture, proving efficient and generalizable but poor in localization and challenged by subtle manipulations.

The main contributions of the proposed research work are:

- A Vision Transformer (ViT)-based model is proposed for image tampering detection and localization, using patch embedding, four transformer encoder layers, and a classification head, achieving strong global feature modeling with only 69,793 trainable parameters.
- The model is evaluated on four standard datasets — CASIA v1, DEFACTO (Splicing), MICC-F2000, and Columbia — covering diverse manipulation types, image formats, and complexities to validate its robustness and generalization.
- Extensive analysis is performed using Accuracy, Precision, Recall, F1 Score, and AUC metrics, ensuring a detailed and consistent validation of the model's detection and localization capabilities across all datasets.
- The proposed method is benchmarked against 59 existing models, consistently outperforming CNN-based, hybrid, and transformer-based architectures across major evaluation metrics on all datasets.
- Detailed visual (confusion matrices, ROC curves) and numerical (metric tables) comparisons confirm the model's superior accuracy, generalization, and scalability in forgery detection and localization.

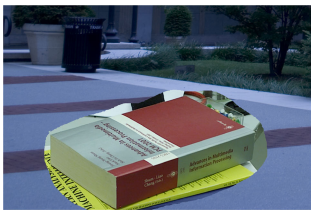
II. DATASET DESCRIPTION

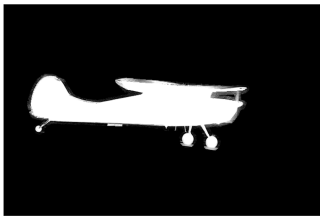
We examined the successful performance of our image forgery detection model utilizing four major datasets: CASIA v1 [14], DEFACTO (Splicing) [23], MICC-F2000 [24], and Columbia [25]. These datasets include both authentic and tampered images. Each dataset moreover supplies ground truth masks that denote the specific tampered areas, enhancing the model's learning of both categorization (authentic or tampered) and localization of alterations.


(a) Authentic CASIA


(b) Tampered CASIA


(c) Authentic Columbia


(d) Tampered Columbia


(e) Authentic DEFACTO splicing


(f) Tampered DEFACTO splicing


(g) Authentic MICC-F2000


(h) Tampered MICC-F2000

FIGURE 1. Examples of authentic and tampered images from the CASIA, Columbia, DEFACTO, and MICC-F2000 datasets.

A. CASIA V1

The CASIA v1 dataset includes 1,721 JPEG images, containing 800 authentic and 921 tampered images. The images are different in size, frequently measuring about 384×256 pixels, and cover both color and grayscale types. The vast majority of tampered images in this dataset are made by splicing, when portions of one image are copied and inserted into another. Refer to Fig. 1a for an example of an authentic image and Fig. 1b for an example of a tampered image

B. DEFACTO (SPLICING)

The DEFACTO dataset contains 984 PNG images, evenly split between authentic and fake. Every image consist of a resolution of 512 × 512 pixels and are uncompressed, producing high-quality data that facilitates the detection of minor or minor modifications. Refer to Fig. 1e for an example of an authentic image and Fig. 1f for an example of a tampered image

C. MICC-F2000

The MICC-F2000 dataset includes 300 high-resolution JPEG photos, containing 130 authentic and 170 tampered images. Image size could reach up to 2048 × 1536 pixels and contain numerous editing types, making this dataset great for testing the model in more complicated forgery scenarios. Refer to Fig. 1g for an example of an authentic image and Fig. 1h for an example of a tampered image

D. COLUMBIA

The Columbia dataset includes 363 TIFF images, containing 183 authentic and 180 tampered images. Every image contain a fixed resolution of 757 × 568 pixels and are uncompressed, which enhances the recognition of finely altered areas. Refer to Fig. 1c for an example of an authentic image and Fig. 1d for an example of a tampered image

E. EVALUATION STRATEGY

We examined our model in two manners: initially, to figure out whether an image is authentic or tampered; and secondly, to recognize the changed areas. We used a standard train-test split and executed cross-dataset evaluation in order to assess the model's generalization capacity on unknown data.

III. PROPOSED MODEL ARCHITECTURE

The entire architecture of the Vision Transformer as shown in Fig. 2 begins with an input RGB image of size 224 × 224 × 3, a standard dimension for image classification tasks. This image is first passed through a Patch Embedding Block, which divides the image into smaller, non-overlapping patches and transforms each patch into a vector.In particular, the image is split into 16 × 16 patches, giving a total of 196 patches (because $224/16 = 14$ and $14^2 = 196$). Each patch is embedded into a 32-dimensional space, creating a matrix of shape 196 × 32, where each row is a patch embedding.

These patch embeddings are then sequentially passed through four Transformer Blocks. Each block captures contextual relationships among patches using self-attention mechanisms. Unlike CNNs that extract features locally, transformers allow global information exchange from the very first layer. After these transformer layers, the output remains 196 × 32, which is then processed by a Global Average Pooling layer. This operation reduces the representation to a single 32-dimensional vector by averaging across all tokens.

The condensed vector is then input into a fully connected classification head consisting of dense layers. This head maps the learned representation to the final output class.In general, the model is based on a modular pipeline of patch embedding, transformer-based feature extraction, and classification.

The input image $I \in \mathbb{R}^{H \times W \times C}$, where $H = W = 224$ and $C = 3$, is partitioned into a grid of fixed-size non-overlapping patches of dimension $P \times P$, with $P = 16$.

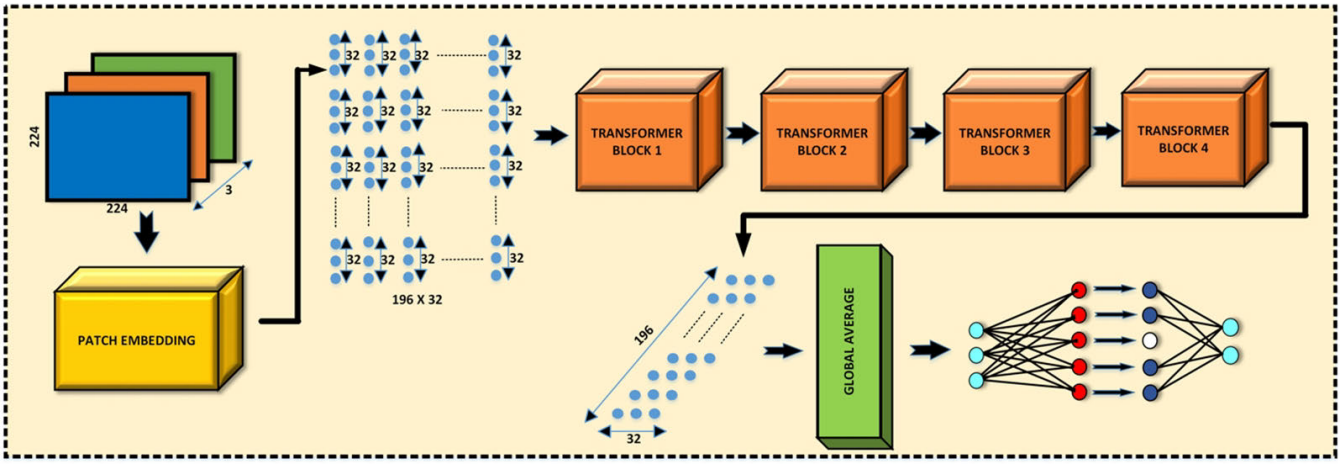


FIGURE 2. Overview architecture of proposed vision transformer model.

The Patch Embedding Block is used to replace the convolutional feature extraction process in conventional CNNs. The input image of dimension $224 \times 224 \times 3$ is split into fixed-size patches of $16 \times 16 \times 3$. This gives us 196 patches, each of which is flattened into a 768-dimensional vector (i.e., $16 \times 16 \times 3 = 768$). These are then fed into a dense layer that projects them into a 32-dimensional latent space to yield the embedding matrix size of 196×32.

A. INPUT REPRESENTATION AND PATCH EMBEDDING

The Patch Embedding Block (Fig. 3) is used to replace the convolutional feature extraction process in conventional CNNs. The input image of dimension $224 \times 224 \times 3$ is split into fixed-size patches of $16 \times 16 \times 3$. This gives us 196 patches, each of which is flattened into a 768-dimensional vector (i.e., $16 \times 16 \times 3 = 768$). These are then fed into a dense layer that projects them into a 32-dimensional latent space to yield the embedding matrix size of 196×32.

This dimensionality reduction is achieved with the preservation of important information and puts data in the form that can be processed by transformers. These embeddings go directly into the transformer encoder. Positional encodings may be appended to these vectors in real-world usage to preserve spatial information, although they are not shown in the diagram.

The number of patches is computed as shown in Eq. (1)

$$N = \frac{H \cdot W}{P^2} = \frac{224^2}{16^2} = 196 \tag{1}$$

Each patch is flattened into a vector $x_i \in \mathbb{R}^{P^2 \cdot C}$, forming a sequence $X = [x_1, x_2, \ldots, x_N] \in \mathbb{R}^{N \times (P^2 \cdot C)}$. A linear projection $W_p \in \mathbb{R}^{(P^2 \cdot C) \times D}$ projects these patch vectors into a lower-dimensional embedding space of dimension $D = 32$ as shown in Eq. (2)

$$z_i = x_i W_p + b_p \quad \text{for } i = 1, \ldots, N \tag{2}$$

The output embeddings $Z = [z_1, \ldots, z_N] \in \mathbb{R}^{N \times D}$ are augmented with a learnable positional encoding $E_{\text{pos}} \in \mathbb{R}^{N \times D}$, resulting as shown in Eq. (3)

$$Z^{(0)} = Z + E_{\text{pos}} \tag{3}$$

This encoding enables the model to retain spatial order information, which is otherwise lost due to the lack of convolution or recurrence.

B. TRANSFORMER ENCODER: MULTI-HEAD SELF-ATTENTION

In the Vision Transformer architecture (Fig. 4), the Transformer Encoder swaps convolutional layers with a attention-based framework. The model contains $L = 4$ stacked encoder layers, every single layer containing two vital modules: Multi-Head Self-Attention (MHSA) and a Position-wise Feedforward Network (FFN). Each module is bound by residual connections and Layer Normalization, creating a residual learning block. This design permits the network to learn about long-range dependencies and global interactions among non-local parts of the input image without depending on the inductive biases fundamental to convolution.

Each Transformer Block comprises two main subcomponents: a Multi-Head Self-Attention (MHSA) mechanism and a Feed-Forward Network (FFN), both enclosed within residual connections and preceded by Layer Normalization. The input to each transformer block is a matrix of shape 196×32.

The first operation is Layer Normalization, normalization across features to improve efficiency and stability. Normalized embeddings are then passed on to the MHSA module, where attention scores over all tokens are calculated such that the model can attend different parts of the image. The attention output is then added to the original input via a residual connection.

Following this, the output undergoes a second Layer Normalization and is passed through a two-layer FFN. The first dense layer expands the feature size from 32 to 64,

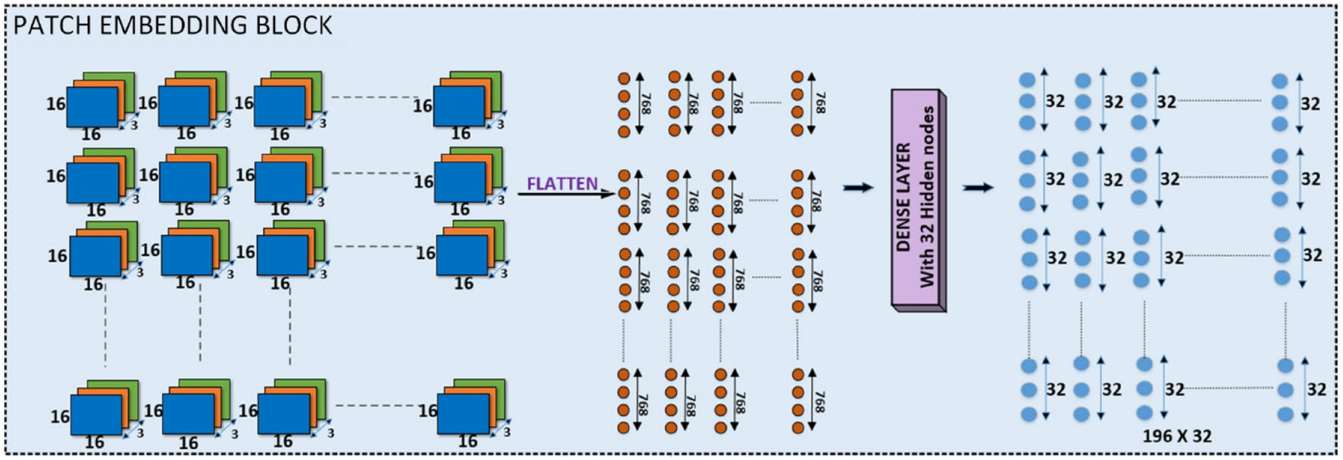


FIGURE 3. Architecture of patch embedding block.

and the second compresses it back to 32, applying nonlinear transformations to each token. Another residual connection feeds the input of this block to its output.

The output still is 196×32 and keeps token structure across different transformer layers. Placing the transformer blocks over each other permits the model to learn more and more abstract features, which relate globally.

1) INPUT EMBEDDING TENSOR

Let the output of the patch embedding module (after position encoding) be denoted as given in Eq. (4)

$$Z^{(l)} = \left[\mathbf{z}_1^{(l)}, \mathbf{z}_2^{(l)}, \ldots, \mathbf{z}_N^{(l)}\right]^\top \in \mathbb{R}^{N \times D} \tag{4}$$

where: Here, $l \in \{0, 1, \ldots, L-1\}$ denotes the current layer index, N is the total number of patches (i.e., the sequence length), D represents the dimensionality of each patch embedding, and $\mathbf{z}_i^{(l)} \in \mathbb{R}^D$ denotes the embedding of the i-th patch at layer l.

2) PROJECTION TO QUERY, KEY, AND VALUE SPACES

Each token vector $\mathbf{z}_i^{(l)}$ is projected into three new representations: Query (q_i), Key (k_i), and Value (v_i) using shared linear projections across the sequence as shown in Eq. (5)

$$Q = Z^{(l)} W_Q, \quad K = Z^{(l)} W_K, \quad V = Z^{(l)} W_V \tag{5}$$

with:

$$W_Q, W_K, W_V \in \mathbb{R}^{D \times d_k}$$

The query, key, and value matrices are $Q, K, V \in \mathbb{R}^{N \times d_k}$, where $d_k = \frac{D}{h}$ is the dimensionality of each attention head, and $h = 2$ is the number of attention heads.

3) SCALED DOT-PRODUCT SELF-ATTENTION

The fundamental attention mechanism computes the affinity between a query q_i and all keys k_j, using the scaled dot-product formula given by equation as shown in Eq. (6)

$$\alpha_{ij} = \frac{\mathbf{q}_i \cdot \mathbf{k}_j^\top}{\sqrt{d_k}} \quad \text{for } i, j = 1, \ldots, N \tag{6}$$

All attention scores are then normalized using the softmax function across each row as given in Eq. (7)

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) \in \mathbb{R}^{N \times N} \tag{7}$$

Each row of A forms a probability distribution over the input tokens. The final attention output for each token is computed as a weighted combination of the values as shown in Eq. (8)

$$\text{head}_i = AV \in \mathbb{R}^{N \times d_k} \tag{8}$$

4) MULTI-HEAD ATTENTION CONCATENATION

To extend the expressiveness of self-attention, h attention heads are used in parallel. Each head learns independent W_Q^i, W_K^i, W_V^i, producing attention outputsc as shown in Eq. (9)

$$\text{head}_i = \text{Attention}(Z^{(l)} W_Q^i, Z^{(l)} W_K^i, Z^{(l)} W_V^i) \quad \text{for } i = 1, \ldots, h \tag{9}$$

The heads are concatenated and projected back to $\mathbb{R}^D$:

$$\text{MHSA}(Z^{(l)}) = \text{Concat}(\text{head}_1, \ldots, \text{head}_h) W_O$$

where $W_O \in \mathbb{R}^{(h \cdot d_k) \times D}$ is a trainable weight matrix for output projection.

5) RESIDUAL CONNECTION AND LAYER NORMALIZATION

To promote stability and ease optimization, the output of the MHSA module is added to its input via a residual connection, and the sum is normalized in the shown Eq. (10)

$$\hat{Z}^{(l)} = \text{LayerNorm}(Z^{(l)} + \text{MHSA}(Z^{(l)})) \in \mathbb{R}^{N \times D} \tag{10}$$

This process ensures that each encoder layer refines the features while preserving the original token representations.

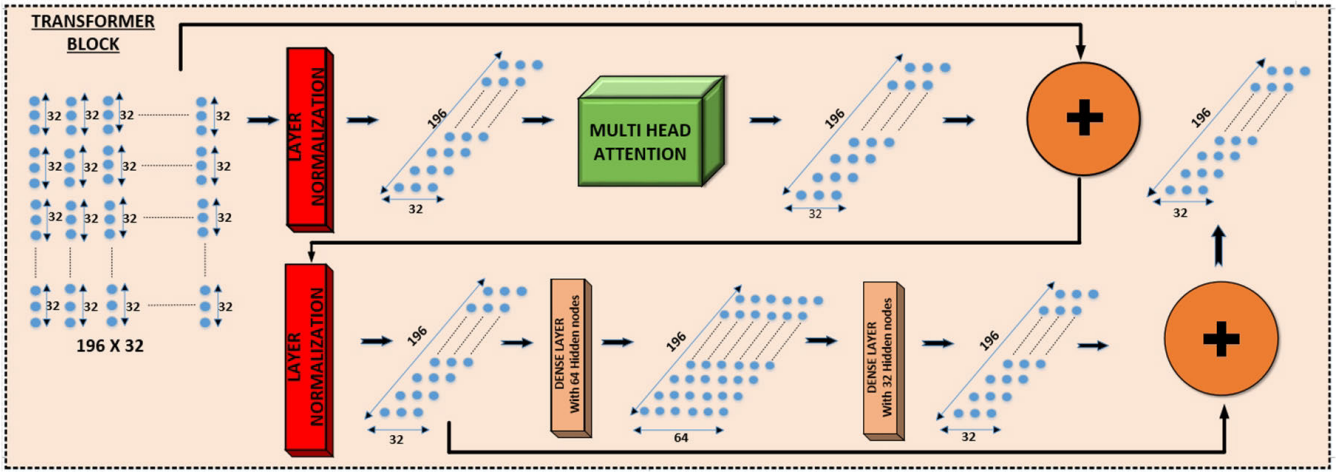


FIGURE 4. Architecture of transformer block.

C. FEEDFORWARD NETWORK AND RESIDUAL LEARNING

After self-attention, the output $\hat{Z}^{(l)}$ is passed through a Position-wise Feedforward Network (FFN). The FFN operates independently on each position $i \in \{1, \ldots, N\}$, applying a two-layer MLP to every token.

1) FEEDFORWARD NETWORK DEFINITION

The FFN function is defined as shown in Eq. (11)

$$\mathrm{FFN}(x_i) = W_2 \cdot \mathrm{ReLU}(W_1 x_i + b_1) + b_2 \tag{11}$$

where: Here, $x_i \in \mathbb{R}^D$ is the i-th row of $\hat{Z}^{(l)}$, $W_1 \in \mathbb{R}^{D\times F}$ and $W_2 \in \mathbb{R}^{F\times D}$ are weight matrices, $F = 64$ represents the hidden (expanded) dimensionality, and $b_1 \in \mathbb{R}^F$, $b_2 \in \mathbb{R}^D$ are learnable bias vectors.

This layer increases the model's capacity to represent complex non-linear transformations.

2) RESIDUAL NORMALIZATION AND OUTPUT OF LAYER

The FFN output is combined with its input through another residual connection and normalized as given in Eq. (12)

$$Z^{(l+1)} = \mathrm{LayerNorm}(\hat{Z}^{(l)} + \mathrm{FFN}(\hat{Z}^{(l)})) \in \mathbb{R}^{N\times D} \tag{12}$$

This marks the conclusion of a fully realized Transformer Encoder layer. Each additional layer processes the output $Z^{(l+1)}$ in a same way, permitting the network to gather progressively abstract and hierarchical representations of the image patches.

D. CLASSIFICATION LAYER AND TRAINING SETUP

Once completion of the final Transformer encoder layer, the output sequence $Z^{(L)} \in \mathbb{R}^{N\times D}$ is compressed into a singular feature vector by the use of global average pooling along the token dimension, resulting as given in Eq. (13)

$$z_{\mathrm{cls}} = \mathrm{GlobalAveragePooling1D}(Z^{(L)}) \in \mathbb{R}^D \tag{13}$$

This consolidated vector encodes a globally contextualized representation of the input image and acts as input to a streamlined classification head. The classification head contains of two entirely linked layers: the very first layer is a dense layer of 128 units, succeeded by a ReLU activation function as shown in the below Eq. (14)

$$h = \mathrm{ReLU}(\mathrm{Dense128}(z\mathrm{cls})) \tag{14}$$

and a dropout regularization with a dropout rate $p = 0.5$ as given in Eq. (15)

$$h = \mathrm{Dropout}(h, 0.5) \tag{15}$$

The second dense layer has a single output neuron with a sigmoid activation function as shown in the below Eq. (16)

$$\hat{y} = \sigma(\mathrm{Dense}_1(h)) = \frac{1}{1 + e^{-h}} \tag{16}$$

which maps the output to a probability score indicating whether the image is tampered ($\hat{y} \approx 1$) or authentic ($\hat{y} \approx 0$). The model is trained using the binary cross-entropy loss function as shown in Eq. (17)

$$\mathcal{L}(y, \hat{y}) = -\left[y \cdot \log(\hat{y}) + (1 - y) \cdot \log(1 - \hat{y})\right] \tag{17}$$

Optimization is conducted with the Adam optimizer with a learning rate $\eta = 5 \times 10^{-5}$, and kernel regularization by weight decay $\lambda = 10^{-3}$ is used in the dense layers to prevent overfitting. The model undergoes training for $E = 80$ epochs with a mini-batch size of $B = 8$, using mixed precision training through the mixed_float16 policy to leverage Tensor Cores for better training efficiency and minimized memory consumption on current GPUs.

IV. RESULTS AND DISCUSSION

In order to evaluate the performance of the proposed Vision Transformer (ViT)-based model for image forgery detection, comprehensive tests were performed on four benchmark datasets: CASIA v1, DEFACTO Splicing, Columbia, and MICC-F2000. Each dataset provided distinct issues regarding resolution, manipulation style, and compression artifacts.

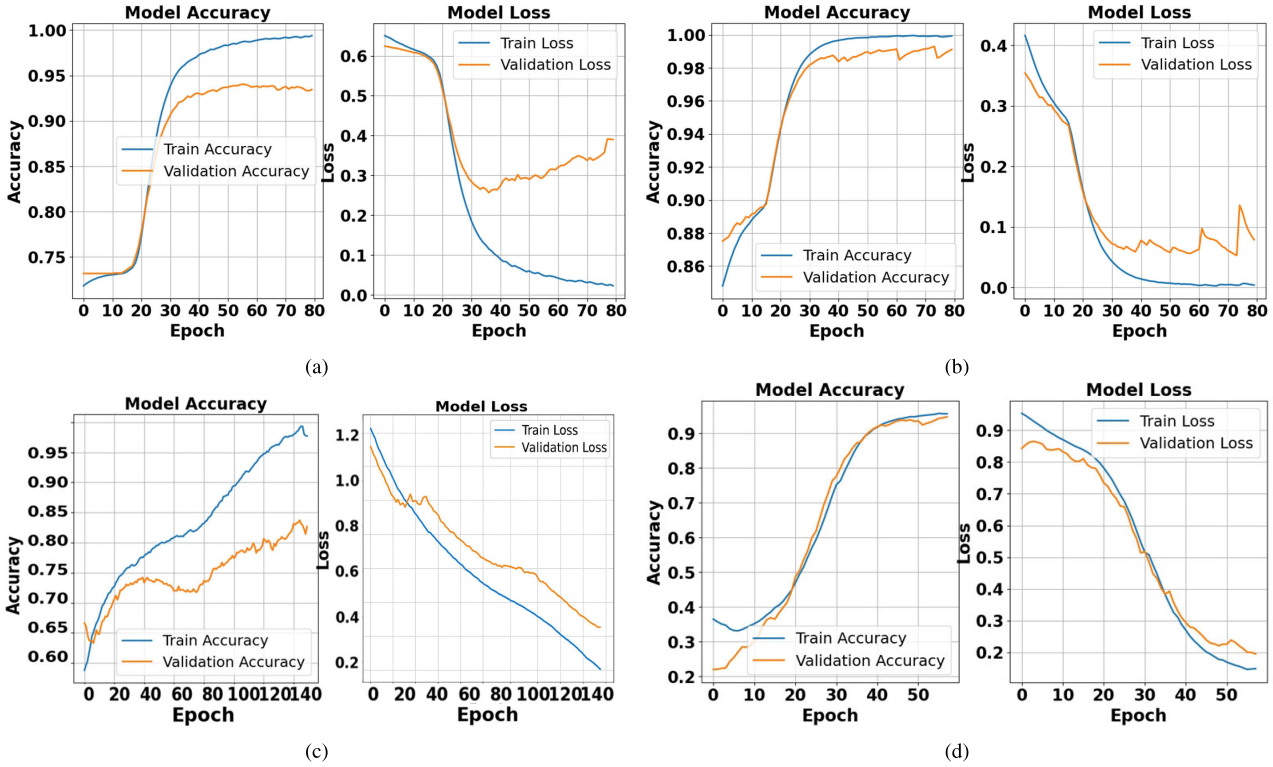


FIGURE 5. Training and validation of model accuracy and model loss graph obtained for (a) CASIA v1 dataset (b) DEFACTO dataset (c) columbia dataset (d) MICC-F2000 dataset by proposed model.

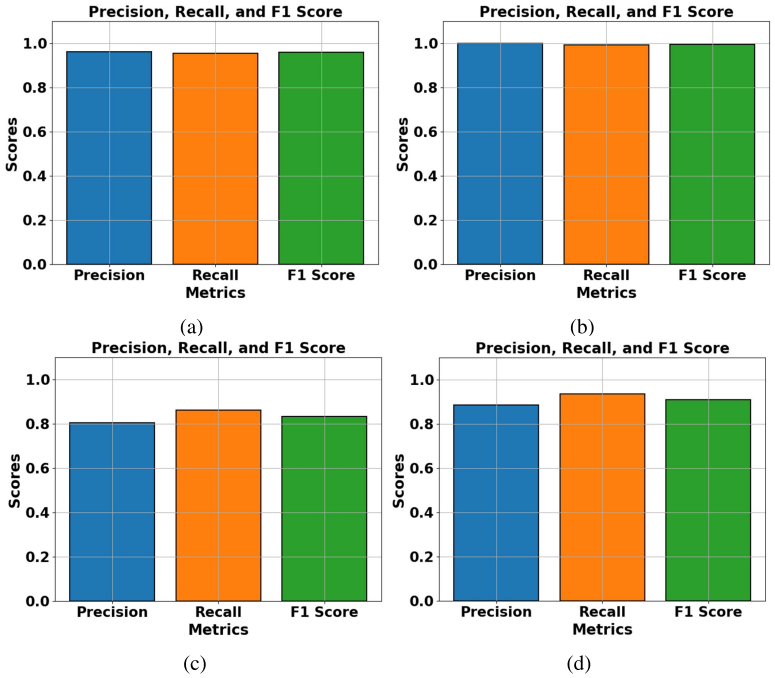


FIGURE 6. Precision, Recall and F1 score obtained for (a) CASIA v1 dataset (b) DEFACTO dataset (c) columbia dataset (d) MICC-f2000 dataset by proposed model.

The model, containing of 69,793 trainable parameters, was individually trained on each dataset for a variable number of epochs (80–200), showing exceptional generalization skills despite its minimal complexity. Performance has been evaluated using standard metrics: Accuracy, Precision, Recall, AUC, and F1 Score.

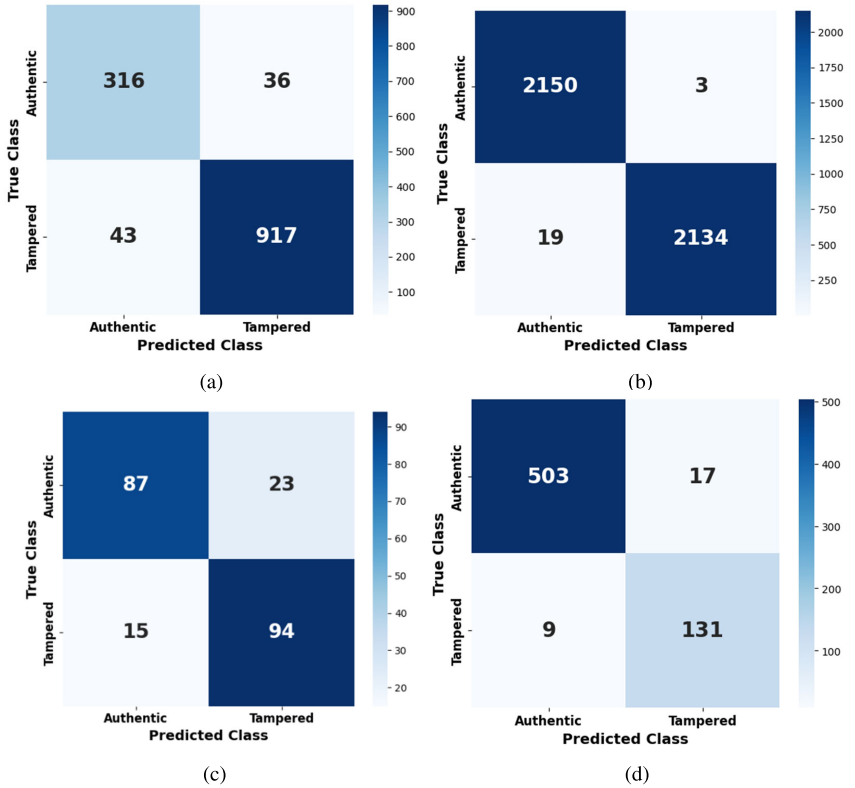


FIGURE 7. Confusion matrix obtained for (a) CASIA v1 dataset (b) DEFACTO splicing dataset (c) columbia dataset (d) MICC-F2000 dataset by proposed model.

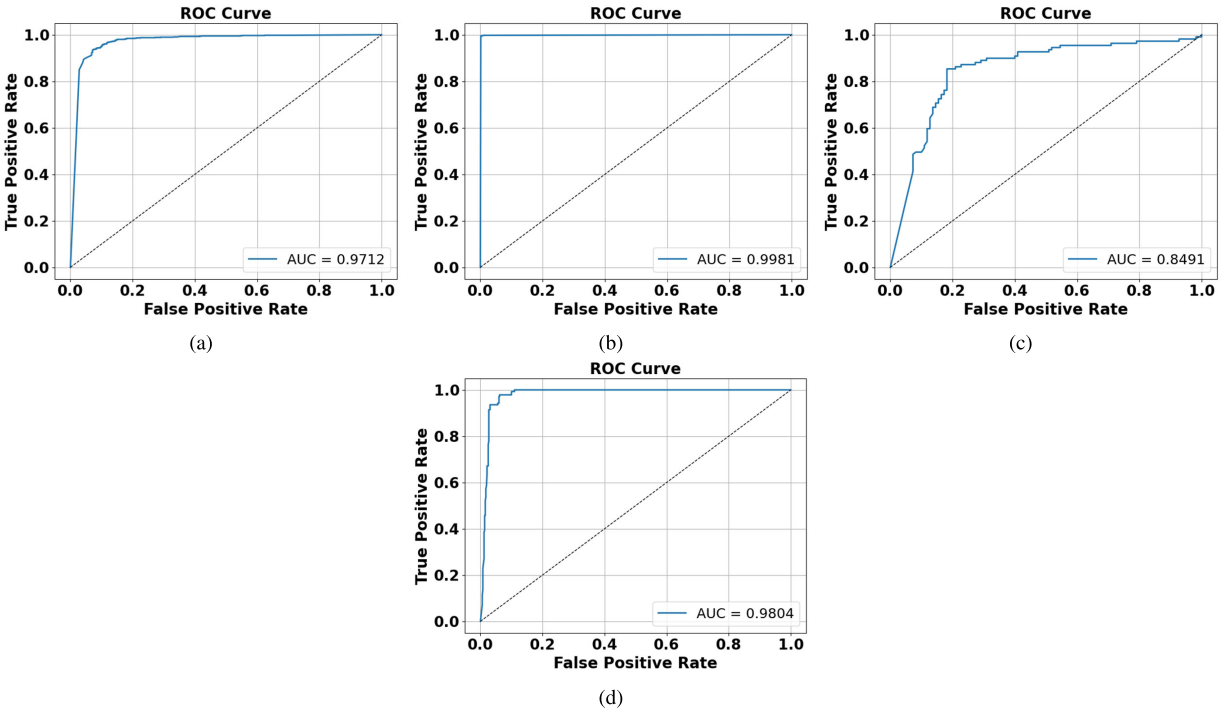


FIGURE 8. ROC curves obtained for (a) CASIA v1 dataset (b) DEFACTO dataset (c) columbia dataset (d) MICC-F2000 dataset by proposed model.

A. ANALYSIS ON CASIA V1 DATASET

The proposed architecture is tested on the CASIA v1 Dataset, which achieved an accuracy of 93.98% as shown in Fig. 5a, precision of 96.22%, recall of 95.52%, and F1 score of 0.9587 as shown in Fig. 6a, and an AUC of 0.9712 as shown in Fig. 8a on the CASIA v1 dataset. These results highlight

the model's equal effectiveness in recognizing splicing-based manipulations. The high AUC implies better differentiation among authentic and tampered samples. CASIA v1 includes both grayscale and color images subjected to lossy JPEG compression, complicating the detection of spliced regions. The ViT's global self-attention mechanism solves this issue by capturing non-local dependencies and spatial inconsistencies that CNNs might miss due to their restricted receptive fields. Moreover, the positional encodings retain spatial connections across patches, making the model attuned to boundary-level aberrations due to splicing. This enables the ViT to accurately identify tampering, even when abnormalities are subtle or spread across distant regions in the image. Furthermore, the confusion matrix for the CASIA v1 dataset, as shown in Fig. 7a, exhibits a strong dominance along the diagonal, indicating a high proportion of correct predictions. Only a very small number of authentic images were misclassified as tampered, and vice versa. This confirms the model's robustness in detecting splicing forgeries even in challenging conditions, demonstrating both high sensitivity and specificity toward forgery detection.

B. ANALYSIS ON DEFACTO SPLICING DATASET

The DEFACTO Splicing dataset showed outstanding results across all benchmarks, with the ViT attaining 99% accuracy as shown in Fig. 5b, 99% precision, 99% recall, and F1 score of 0.99 as shown in Fig. 6b, and an AUC of 0.99 as shown in Fig. 8b. The nearly perfect outcomes illustrate the model's resilience in analyzing uncompressed, high-resolution images with modest splicing manipulations. This dataset's accurate preservation of low-level textures and lack of compression noise allowed the ViT to use extensive token-wise feature representations. The multi-head focus system allows the collection of various representations of spatial relationships, assisting the model in accurately identifying manipulated areas. In addition, training for 80 epochs certainly helped the model to refine attention weights and positional encodings, thereby optimizing both detection sensitivity and specificity. The results obtained demonstrate the ViT's ability to be used as a high-resolution forgery detector in optimal image conditions. Moreover, the confusion matrix for the DEFACTO Splicing dataset, as illustrated in Fig. 7b, displays almost perfect diagonal dominance, with minimal false positives and false negatives. The extremely low rate of misclassification highlights the model's precision in distinguishing authentic images from tampered ones. This strong performance confirms the model's capability to accurately capture and localize fine-grained manipulations even in high-quality, minimally altered images.

C. ANALYSIS ON COLUMBIA DATASET

The Columbia dataset presented a more significant issue owing to its uncompressed TIFF format and reduced dataset size. The ViT model, after 150 epochs of training, achieved an accuracy of 82.65% as shown in Fig. 5c, precision of 80.34%, recall of 86.24%, and F1 score of 0.8319 as shown in Fig. 6c, and an AUC of 0.8491 as shown in Fig. 8c. Despite being significantly lower than in other datasets, these values remain competitive and significantly outperform the performance of many previous methods. The model's comparatively higher recall implies robust sensitivity for detecting manipulated images, though the marginally reduced precision indicates occasional misclassification of authentic samples. This is because of a lack of diversity of augmentations in the dataset and the decreased rate of explicit manipulation artifacts. Still, the transformer's capability to build long-range dependencies among uniform image backgrounds permits it to outperform traditional CNN-based models. The extended training duration certainly reduced the dataset's size limitations by permitting more refined tuning of inter-patch interactions and token embedding weights. In addition, the confusion matrix for the Columbia dataset, shown in Fig. 7c, reflects a modest increase in misclassifications compared to other datasets, with some authentic images incorrectly labeled as tampered and vice versa. However, the majority of the samples were still correctly classified, and the diagonal dominance was maintained, confirming the ViT model's general capability to handle challenging, less diverse datasets with minimal overfitting.

D. ANALYSIS ON MICC-F2000 DATASET

The MICC-F2000 dataset contains high-resolution images exhibiting several forgery techniques. The ViT model, trained for 200 epochs, obtained an accuracy of 96.06% as shown in Fig. 5d, precision of 88.51%, recall of 93.57%, and F1 score of 0.9097 as shown in Fig. 6d, and an AUC of 0.9804 as shown in Fig. 8d. These results are particularly significant due to the dataset's inherent complexity, which includes different manipulation strategies and subtle semantic inconsistencies. The high recall implies that the model successfully detects a variety of tampering types, including both overt and covert manipulations. The application of global average pooling and regularization techniques (dropout and L2 kernel penalties) prevents overfitting, even with an extended training duration. The model's solid performance with a comparatively low parameter count shows the effectiveness of transformer encoders in extracting and retaining hierarchical tampering clues. The global field-of-view given by ViTs provides enhanced feature aggregation, which is especially important for detecting distributed or context-aware forgeries in MICC-F2000. Furthermore, the confusion matrix for the MICC-F2000 dataset, presented in Fig. 7d, shows a strong concentration along the diagonal, indicating a high proportion of correct classifications. A few misclassifications are observed, mostly between subtle tampering and authentic images, which is expected given the dataset's complex and fine-grained manipulations. Overall, the confusion matrix validates the model's robustness and its ability to generalize effectively across varying types of forgery techniques in high-resolution images.

The consistent excellent performance within all datasets is due to the basic benefits of the Vision Transformer architecture: its capability to model long-range interrelationships, its represented input representation, and its adaptability in learning complicated spatial patterns without dependence on handcrafted features or convolutional filters. Despite including over $\sim$ 69.7K parameters, the model showcases state-of-the-art performance, resulting in both accurate and lightweight. Mixed precision training improves computational efficiency and permitting deployment in resource-constrained environments. The ViT generalizes effectively among datasets with various resolutions, formats, and complexities by refusing convolutional inductive biases in favor of attention-based feature alignment.

The porposed model taken inference time of 6.2 seconds for testing of 4306 forgered images each of size $224 \times 224 \times 1$ in batch wise manner with batch size of 32.

E. ABLATION STUDY

The ablation study shown in the Fig. 9 clearly demonstrates that the proposed model, using 4 Transformer Blocks (TB's), strikes an excellent balance between performance and computational efficiency. As the number of Transformer Blocks increases, the number of parameters increases significantly, leading to higher computational costs. However, the proposed model (4 TB's) achieves high accuracy and AUC while maintaining a relatively low number of parameters, reaching a sweet spot at 4 TB's. This is evident as the model with 4 TB's achieves an accuracy of 99.49% and an AUC of 99.19% with 69.7K parameters, making it a highly efficient solution compared to models with 5 TB's, where the accuracy and AUC increase marginally, but the number of parameters increases to 82.5K. This suggests that the model's performance plateaus beyond 4 TB's, which highlights the diminishing returns in terms of accuracy with the increase in Transformer Blocks. The proposed model with 4 TB's is an effective approach to achieving high performance while keeping the model size and computational cost manageable.

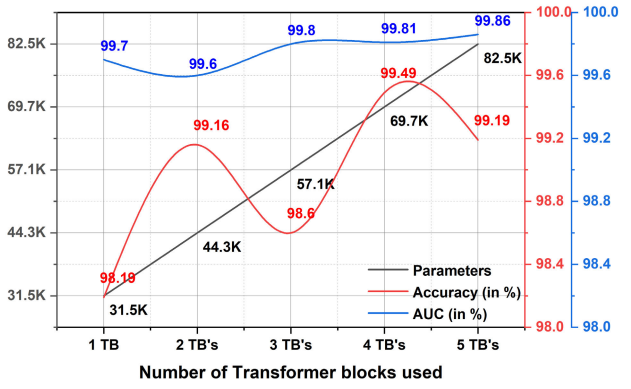


FIGURE 9. Performance comparison of the proposed model (4 Transformer Blocks) showing accuracy, AUC, and parameter efficiency.

Further, a ablation study shown in Fig. 10 analyzes the effect of different patch sizes (16×16, 8×8, 32×32, and 4×4) on the performance of the model, in terms of accuracy, AUC (Area Under Curve), and the number of parameters. From the plot, it is evident that as the patch size decreases from 16×16 to 4×4, the number of parameters increases significantly. This trend suggests that smaller patches require more parameters to capture the information across the image, leading to increased model complexity. On the other hand, larger patch sizes like 32×32 lead to a decrease in accuracy and AUC, indicating that using too large of a patch size reduces the model's ability to capture fine-grained details, which are crucial for performance.

The proposed work used a 16×16 patch size which strikes an optimal balance, achieving high accuracy (99.81%) and AUC (99.44%) with only 69.7K parameters. This highlights the efficiency of using the 16×16 patch size in the proposed work, where good performance is achieved.

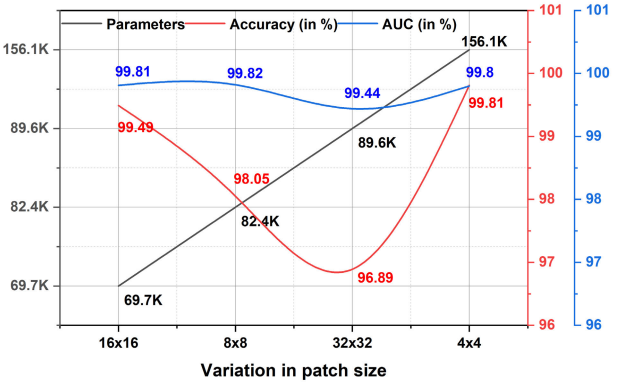


FIGURE 10. The effect of patch size variation on accuracy, AUC, and parameters, with the proposed model using a 16 × 16 patch size.

V. QUALITATIVE AND QUANTITATIVE COMPARATIVE ANALYSIS

The proposed ViT model shows extraordinary superiority against various existing state-of-the-art models across the CASIA v1, DEFACTO, Columbia, and MICC-F2000 datasets. On CASIA v1, ViT obtains the highest AUC (0.9712) and F1 score (0.9587), greatly outperforming existing models such as CNN+DCT (F1 = 0.95) [26], ForensicNet (F1 = 0.92) [27], and MVSS-Net (F1 = 0.75) [28]. Recent hybrid transformers such as IML-ViT [29] and VIT-132 [9] have performance levels approaching that of ViT (F1 = 0.81 and 0.94, respectively); however, they are low in generalization and recall metrics. ViT significantly outperforms lightweight networks like ISIE-Net (F1 = 0.53) [30] and MAPS-Net (F1 = 0.55) [31], as well as deeper localization models such as TransForensics and TDA-Net, having F1 values below 0.63 (Table. 1). The performance is attributed to ViT's architecture, it captures non-local dependencies with self-attention and encodes spatial continuity through learnable positional embeddings, which resulted in major resilience to texture variance and compression flaws found in CASIA v1.

TABLE 1. Performance metrics comparison of various models on CASIA v1 dataset from (2020–2025).

Ref.NO	Model	Year	AUC	F1 Score	Accuracy	Precision	Recall
[14]	CANF	2024	0.91	0.73	-	-	-
[32]	CCLHF-Net	2024	0.84	0.75	-	-	-
[26]	CNN + DCT	2020	-	0.95	0.90	0.96	0.94
[11]	ConvNeXtFF	2023	0.85	0.58	-	-	-
[33]	CR-CNN	2020	0.78	0.47	-	-	-
[34]	EITL-Net	2024	-	0.55	-	-	-
[35]	EMT-Net	2023	0.85	0.45	-	-	-
[36]	EXIF-SC	2020	-	0.61	-	-	-
[37]	FCN	2021	0.79	0.77	-	-	-
[27]	Forensic-Net	2023	-	0.92	-	0.90	0.94
[10]	GSC-Net	2020	0.83	0.47	-	-	-
[38]	GSR-Net	2020	-	0.57	-	-	-
[39]	HiFi-Net	2023	-	0.09	-	-	-
[40]	IF-OSN	2022	0.87	0.5	-	-	-
[29]	IML-ViT	2024	-	0.81	-	-	-
[30]	ISIE-Net	2025	0.82	0.53	-	-	-
[15]	locate-Net	2022	0.75	0.27	-	-	-
[15]	Loma	2025	-	0.76	-	-	-
[41]	M2BG	2024	-	0.65	-	-	-
[31]	MAPS-Net	2024	0.8	0.55	-	-	-
[42]	MPC	2025	-	0.74	-	-	-
[43]	MSFF	2022	0.7	0.45	-	-	-
[44]	MSF-Net	2025	0.9	0.63	-	-	-
[28]	MVSS-Net	2022	0.93	0.75	-	-	-
[28]	MVSS-Net++	2022	-	0.77	-	-	-
[45]	NCL	2023	0.86	0.59	-	-	-
[46]	NCNS	2023	0.85	0.66	-	-	-
[47]	OSN	2022	0.87	0.5	-	-	-
[48]	PCL	2023	0.75	0.46	-	-	-
[49]	PSCC-NEt	2023	0.87	0.55	-	-	-
[43]	RRA-Net	2022	-	0.7	-	-	-
[44]	SPAN	2020	0.83	0.38	-	-	-
[50]	TB-Net	2023	-	0.59	-	-	-
[51]	TDA-Net	2021	0.83	0.58	-	-	-
[52]	TransForensics	2021	0.83	0.62	-	-	-
[53]	TruFOR	2023	-	0.69	-	-	-
[13]	VSS+SS2D	2025	-	0.72	-	-	-
[9]	VIT-132	2025	-	0.94	-	0.91	0.90
Proposed model		2025	0.9712	0.9587	0.9398	0.9622	0.9552

TABLE 2. Performance metrics comparison of various models on DEFACTO splicing dataset.

Ref.No	Model	Year	AUC	F1 Score	Accuracy	Precision	Recall
[23]	CAT-Net	2023	-	-	0.99	-	-
[23]	DRRU-Net	2023	-	-	0.99	-	-
[23]	DRRU-Net LR	2023	-	-	0.99	-	-
[23]	DRRU-Net TL	2023	-	-	0.99	-	-
[23]	RRU-Net	2023	-	-	0.988	-	-
Proposed model		2025	0.9993	0.9914	0.9914	0.9921	0.9907

In the DEFACTO dataset, characterized by high-resolution, uncompressed images and subtle splicing forgeries, ViT obtains almost perfect results: AUC = 0.9993, F1 = 0.9914, and accuracy = 99.14%, outperforming all evaluated models, including CAT-Net and DRRU-Net [23], which provide only accuracy metrics without AUC or F1 scores (Table.2). ViT keeps an equal amount of precision (99.21%) and recall (99.07%), which is essential for minimising false positives in practical applications.

In the Columbia dataset, ViT surpasses models such as C2R-Net (F1 = 0.69) [54], EXIF-SC (F1 = 0.51) [36], and DFCN (F1 = 0.41) [25], and equals to HiFi-Net (F1 = 0.83) [39] while showing superior recall (0.86) (Table.3). The effectiveness of the model in Columbia, although its low image diversity and organized format, is due to ViT's attention-based global feature modeling, which reliably recognizes semantic-level tampering often overlooked by convolutional models limited by local receptive fields.

TABLE 3. Performance metrics comparison of various models on Columbia dataset.

Ref.	Model	Year	AUC	F1 Score	Accuracy	Precision	Recall
[54]	C2R-Net	2020	-	0.69	-	0.80	0.61
[55]	CAT-NET	2021	-	0.55	-	-	-
[56]	CAT-NET	2022	-	0.79	-	-	-
[33]	CR-CNN	2020	-	0.43	-	-	-
[25]	DFCN	2021	0.62	0.41	-	-	-
[36]	EXIF-SC	2020	-	0.51	-	-	-
[37]	FCN	2022	-	0.53	-	-	-
[27]	ForensicNet	2023	-	0.63	-	0.65	0.62
[38]	GSR-Net	2020	-	0.61	-	-	-
[39]	HiFi-Net	2023	-	0.83	-	-	-
[44]	MSF-Net	2025	-	0.78	-	-	-
[57]	Noiseprint	2020	0.84	0.36	-	-	-
[47]	PCL	2023	0.76	0.69	-	-	-
[49]	RRA-Net	2022	-	0.75	-	-	-
[43]	SPAN	2020	-	0.48	-	-	-
[50]	TruFOR	2023	-	0.79	-	-	-
Proposed model		2025	0.84	0.83	0.82	0.80	0.86

ViT overcomes hybrid CNN-transformer architectures which include MSF-Net, GSR-Net, and PSCC-Net, which display acceptable F1 scores (∼0.6–0.7) but demonstrate insufficient robustness in cross-context manipulation.

The MICC-F2000 dataset, containing high-resolution images exhibiting manipulation techniques, significantly confirms the adaptability of the ViT model. Having an AUC of 0.9804, an F1 score of 0.9097, and an accuracy of 96%, ViT surpasses conventional frequency-based techniques like as DCT+SVM and ELA+VGG16 [24] which show poor generalization(Acc = 93% and 89%, respectively) (Table.4). In comparison to CNNs, which depend on spatial localization or frequency artifacts, ViT has better ability in modeling both global and local tampering patterns with its tokenized patch attention and hierarchical feedforward layers. In summary, ViT outperforms convolutional architectures (CR-CNN, ConvNeXtFF, MSF-Net), competitive and contrastive methods (TruFOR, locate-Net, NCL), and even advanced transformer variants (TransForensics, IML-ViT), by consistently showing superiority across all five evaluation metrics (Accuracy, Precision, Recall, AUC, F1). With only 69,793 trainable parameters, four encoder layers, and effective mixed-precision training, ViT shows itself as a high-performing and lightweight architecture—establishing a new benchmark for dependable and scalable imagine tampering detection across various datasets.

VI. CONCLUSION

In this work, we proposed a lightweight Vision Transformer-based architecture for image tampering detection and localization, addressing the growing need for efficient and scalable

TABLE 4. Performance metrics comparison of various models on MICC-F2000 dataset.

Ref.No	Model	Year	AUC	F1 Score	Accuracy	Precision	Recall
[24]	DCT+SVM	2023	-	-	0.93	0.88	0.9
[24]	ELA+VGG16	2023	-	-	0.89	0.88	0.9
[24]	ELA+RESNET50	2023	-	-	0.76	0.65	0.8
Proposed model		2025	0.98	0.90	0.96	0.88	0.93

forgery detection solutions in digital forensics. By leveraging patch embedding, multi-head self-attention, and a modular classification pipeline, the proposed model effectively captures global feature representations while maintaining a compact design with only 69,793 trainable parameters. Comprehensive experiments conducted on four benchmark datasets—CASIA v1, DEFACTO (Splicing), Columbia, and MICC-F2000—demonstrate the robustness, adaptability, and high accuracy of the proposed approach across diverse manipulation scenarios and image types. The model achieves impressive results, including an F1 score of 0.9587 and an AUC of 0.9712 on CASIA v1, an almost perfect F1 score of 0.9914 on DEFACTO, and strong performance even on challenging datasets like Columbia and MICC-F2000. Extensive qualitative and quantitative comparisons against 59 existing state-of-the-art models further validate the superior performance of the proposed ViT model in terms of accuracy, precision, recall, AUC, and F1 score across all datasets. The results clearly establish the proposed model as a high-performing, lightweight, and scalable solution for real-world image tampering detection and localization applications. This work distinguishes itself from prior ViT-based methods by demonstrating that a compact transformer architecture can deliver state-of-the-art performance across diverse forgery datasets, while significantly reducing model complexity and computational cost. Future directions include incorporating data augmentation, transfer learning, and multi-scale learning to enhance the model's robustness and generalization across diverse and challenging datasets.

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